JASON LEE

linkedin.com/in/jibrailjasonlee — jjason.lee@mail.utoronto.ca — jibrailjasonlee.github.io

EDUCATION

University of Toronto Sept 2025 - Present
Bachelors in Applied Sciences in Mechanical Engineering - GPA: 3.74 / 4.00 Toronto, ON

PROJECTS

Client-Based Vertical Hydroponic Tower Redesign Jan 2026 – Apr 2026
Project Manager *Toronto, ON*

- Designed a hydroponic tower using SOLIDWORKS and Fusion 360, managing a 6-person team through design development, CAD modeling, and subsystem integration.
- Fabricated 48 custom PLA net cups via rapid 3D printing, optimizing geometry and seal interfaces to eliminate leakage.
- Simulated drainage performance in MATLAB and Excel, validating a 5,544 mm^2 discharge area through fluid mechanics modeling to prevent volumetric overflow.
- Prototyped and stress-tested subsystem components, achieving 0 mL leakage through iterative design refinement and hydrostatic load testing.

Piston Machining Sept 2026
George Brown College Machining Course *Toronto, ON*

- Machined a functional piston to engineering drawing specifications over two intensive weekend sessions, executing manual lathe turning, facing, and mill-based operations on raw stock.
- Partnered with 1 teammate to interpret blueprint dimensions and GD&T callouts, sequencing multi-step setups across the lathe, mill, and machine vise to maintain specified tolerances.
- Verified dimensional accuracy and surface finish using precision metrology tools, applying workholding and fixturing techniques to ensure part conformance under time-constrained shop conditions.

LEGO Mindstorms EV3 2021
Independent Project *Hong Kong*

- Built an autonomous EV3 robot using block-based logic, implementing motor control, basic kinematics, and sensor-driven navigation routines.
- Prototyped and refined mechanical subsystems through iterative testing, improving structural stability and task execution reliability.

WORK EXPERIENCE

Technical Production Lead Nov 2024 - Present
Hoop City Basketball League *Toronto, ON*

- Optimized digital production pipelines for 30+ seasonal events, maintaining a 100% delivery rate for high-precision technical assets within strict 24-hour windows.
- Analyzed engagement metrics using Excel-based data modeling to drive content optimization, resulting in a 200% increase in monthly viewership and 5,000+ new followers.
- Managed high-value digital hardware and assets, ensuring operational readiness for live sports environments.

Head Photographer & Content Manager Oct 2023 - June 2025
Monarch Park Collegiate Institute *Toronto, ON*

- Directed a technical team of 6 photographers, overseeing quality assurance and metadata management for an institutional database of 5,000+ digital assets.
- Mentored junior staff in hardware maintenance protocols and professional post-processing workflows, increasing team efficiency and deliverable fidelity.

UNIVERSITY INVOLVEMENT & CONTINUED LEARNING

University of Toronto Robotics Association – Autonomous Rover Team Sept 2026 – Present
Member, Mechanical Subteam *Toronto, ON*

· Designing a rover chassis and drivetrain in SOLIDWORKS for the 2027 Intelligent Ground Vehicle Competition, researching motor selection and T-slot framing to meet performance and budget requirements.

Robotics for Space Exploration – Rover Arm Team Sept 2026 – Present
Member, Mechanical Subteam *Toronto, ON*

· Designing a camera mount with an integrated pan-tilt mechanism in Onshape to enable a controllable webcam on the rover chassis.

SKILLS AND TOOLS

Technical Skills: SOLIDWORKS, Fusion 360, Onshape, PLA 3D Printing, Machining
Robotics & Embedded Systems: Python, C/C++, Microcontrollers, Kinematics, Sensor Integration
Analysis & Simulation: MATLAB, Simulink, Minitab